MILITARY SPECIFICATION SHEET

ELECTRON TUBE, NEGATIVE GRID (MICROWAVE)

1/ TYPE 8980

This specification is approved for use by Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Triode, planar, ceramic and metal construction
See figure 1
Mounting position: Any
Weight: 0.5 ounce (14 grams) nominal

ABSOLUTE RATINGS:

Parameter: Unit:	F GHz	Ef V	Eb Vdc	Ec Vdc	Ik mAdc	Ib mAdc	Ic mAdc	Pp W	Pg W	tk sec (min)	TE & anode shank T °C (Note 1)	Cooling --- (Note 1)
Maximum: CW osc or amplifier:	3.0	6.0 ±2%	500	-50	35	35	5	10	0.5	300	250	---
TEST CONDITIONS:	---	6.0	180	Adj	---	---	---	---	---	300	---	Note 2

GENERAL:

Qualification - Required
Vaporization shields - Required (See note 3)

1/ Replaces EIMAC Y-638 and 6503

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1	4							
1301	Heater current	-		0.65	II	If	530	590	mA
---	Constant grid voltage	5		0.65	II	---	---	---	---
	Quality conformance inspection, part 2								
1256	Electrode current (1) (anode)	-	Ec = -6.0 Vdc	---	---	Ib	6.0	40.0	mAdc
1256	Electrode current (2) (anode)	-	Eb = 75 Vdc; Ec = Ek = 0 Vdc	---	---	Ib	10.0	40.0	mAdc
1261	Electrode voltage (grid)	-	Eb = 150 Vdc; Ec/Ib = 15 mAdc	---	---	Ec	-2.0	-7.5	Vdc
1316	Amplification factor	-	Eb = 75 Vdc; Ec = 0 Vdc	---	---	Mu	14.0	28.0	---
1306	Transconductance	-	Eb = 75 Vdc; Ec = 0 Vdc	---	---	Sm	2,500	10,000	µmhos
1331	Direct-interelectrode capacitance (grounded cathode connection)	6, 9		---	---	Cgk Cgp Cpk	1.30 1.95 ---	1.70 2.35 0.04	pF pF pF
1236	Power oscillation	7	Ebb = 150 Vdc; Rk/Ik = 25 mAdc	---	---	Po	20.0	---	mW
---	Cold resonance	8, 9		---	---	---	---	---	---
	Quality conformance inspection, part 3								
---	Life test	-	Group C; heater standby Ef = 6.0 V; no other voltages applied; t = 500 hours	---	---	---	---	---	---
---	Life-test end point:								
1236	Power oscillation	-		---	---	Po	15.0	---	mW

NOTES:

1. Sufficient conduction, convection, or forced-air cooling must be provided for the anode, cathode shank, and all seals to limit the maximum temperature to the specified value under all operating conditions. Where emphasis is placed on long life and consistent performance cooling in excess of minimum requirements is normally beneficial.
2. During all electrical testing involving application of heater voltage sufficient cooling may be used to assure operation within the specified maximum temperature rating.
3. Internal vaporization shields are required and shall be arranged so that no vaporization products are deposited on critical areas of the inner surface of the ceramic envelope.
4. All tests listed under quality conformance inspection, part 1 shall be performed at the conclusion of the holding period.
5. Constant grid voltage tests will be performed by means of a curve tracer. Anode voltage (eb) shall vary between 0 and 180 v. Limits for constant grid voltages are defined as follows:

eb	Grid voltage = 0 V		Grid voltage = -6 V	
	ib = ma (min)	ib = ma (max)	ib = ma (min)	ib = ma (max)
75 v	10.0	40.0	---	---
150 v	---	---	2.0	33.0
180 v	30.0	80.0	6.0	40.0

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NOTES: -Continued

6. Measure in test fixture per DESC Drawing D76025.

7. Measure in stalo cavity (NSN 5840-00-522-3019) in accordance with figure 2. The output coupling loops (2) shall be set and locked at the 45° position. Anode supply shall be a well regulated 150 Vdc (±5%) supply and heater supply shall be a well regulated 6.0 Vdc (±0.1 V) supply. The anode current shall be adjusted to be within 20 to 30 mAdc by adjustment of an external cathode bias resistor. Output from one of two coupling loops of the stalo assembly shall be 20.0 mW minimum. All tuning adjustments are to be used to the fullest extent to make the tubes operate in accordance with the requirements. With the AFC control set in the electrical center of its range, and with all tuning rods locked in position, the power will be measured at 2.67, 2.80, and 2.93 GHz, observing that throughout the tuning range between the limit frequencies the power shall not drop below 20 mW. The maximum cathode current shall not at any time exceed 40 mAdc.

8. Grid-anode resonance. Test in cavity in accordance with DESC Drawing D76024. Cavity shall resonate at 2,022 ±2 MHz with tuning slug in accordance with DESC Drawing C76022 at TA = 25° ±5°C.

   Grid-cathode resonance. Test cavity in accordance with DESC Drawing C76026. Cavity shall resonate at 4,420 ±5 MHz with tuning slug in accordance with DESC Drawing C76022.

   When plotted on graphs of resonant frequency versus grid-anode capacitance and resonant frequency versus grid-cathode (input) capacitance, the tube under test shall be represented by a point within a parallelogram whose four corners are located by the following points:

Points	Capacitance (pF)		Frequency (MHz)	
	Cgp	Cgk	Fgp	Fgk
1	1.95	1.3	1,252	1,386
2	1.95	1.3	1,282	1,446
3	2.35	1.7	1,224	1,264
4	2.35	1.7	1,254	1,324

9. All DESC drawings are available from Defense Electronics Supply Center, DESC-ES, Dayton, Ohio 45444.

User activities:
Navy - AS, OS, MC, CG

Preparing activity:
Navy - EC

Agent:
DLA - ES

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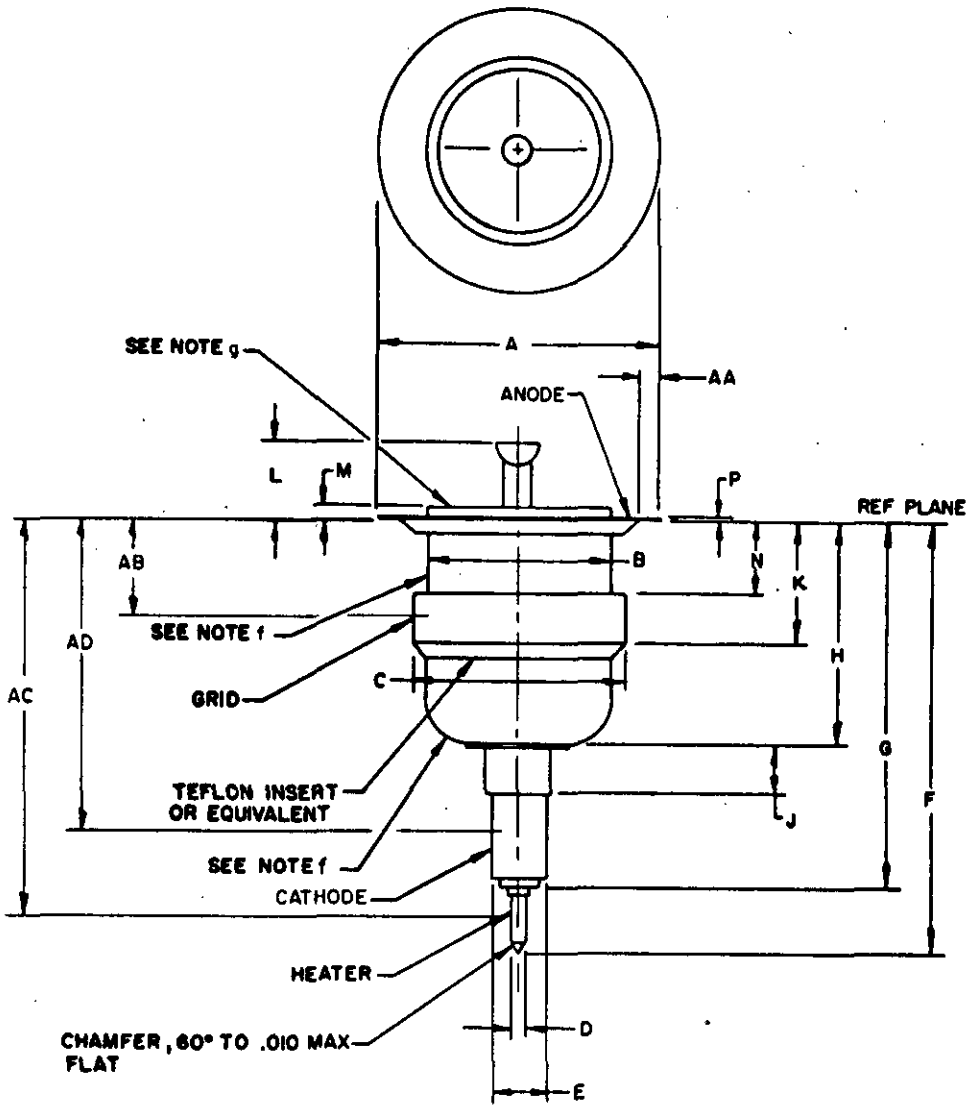


FIGURE 1. Outline drawing of electron tube type 8980.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses		Notes
	Minimum	Maximum	
Quality conformance inspection, part 2			
A	.990 (25.15)	1.010 (25.65)	b
C	.740 (18.80)	.750 (19.05)	b
D	.047 (1.19)	.053 (1.35)	b
E	.197 (5.00)	.207 (5.26)	b
F	1.370 (34.80)	1.430 (36.32)	
G	1.150 (29.21)	1.205 (30.61)	
K	.365 (9.27)	.435 (11.05)	
L	.750 (19.05)	.850 (21.59)	
Quality conformance inspection, part 3			a
B	.640 (16.26)	.660 (16.76)	
H	.675 (17.14)	.775 (19.68)	
J	.125 (3.18)	.195 (4.95)	
M	.040 (1.02)	.075 (1.90)	
N	.200 (5.08)	.270 (6.86)	
P	.007 (.18)	.013 (.33)	
Electrode contact dimensions			e
AA	.080 (2.03)		c,d
AB	.275 (6.98)	.360 (9.14)	c,d
AC	1.230 (31.24)	1.340 (34.04)	d
AD	.980 (24.89)	1.100 (27.94)	d

NOTES:

a. These dimensions shall be tested monthly when in continuous production, with sampling as follows:

$n_1 = 4$ $c_1 = 0$

$n_2 = 4$ $c_2 = 1$; where c_2 represents the total allowable failures for the first and second samples combined.

In case of failure after double sampling, the failing dimension(s) shall become a part of quality conformance inspection, part 2, AQL 6.5, inspection level S3. After three consecutive successful submissions the testing shall revert to the monthly basis.

b. Gold plate (note a applies).

c. Eccentricity of contact surfaces shall be gaged from the centerline of reference and shall be as follows (note a applies).

Contact surface	TIR (Max)	Reference
Anode (A)	.040 (1.02)	Cathode (E)
Grid (C)	.020 (.51)	Cathode (E)

d. Dimensions of contact surfaces shall apply throughout entire contact area.

e. Dimensions in electrode contact area table are for quality conformance inspection, part 2 purposes only, and are not intended for inspection purposes.

f. This surface shall not be used for clamping or locating.

g. Eccentricity of backup ceramic to anode (A) .031 (.79 mm) maximum; to be measured with EIMAC gage G-2285, or equivalent.

FIGURE 1. Outline drawing of electron tube type 8980 - Continued.

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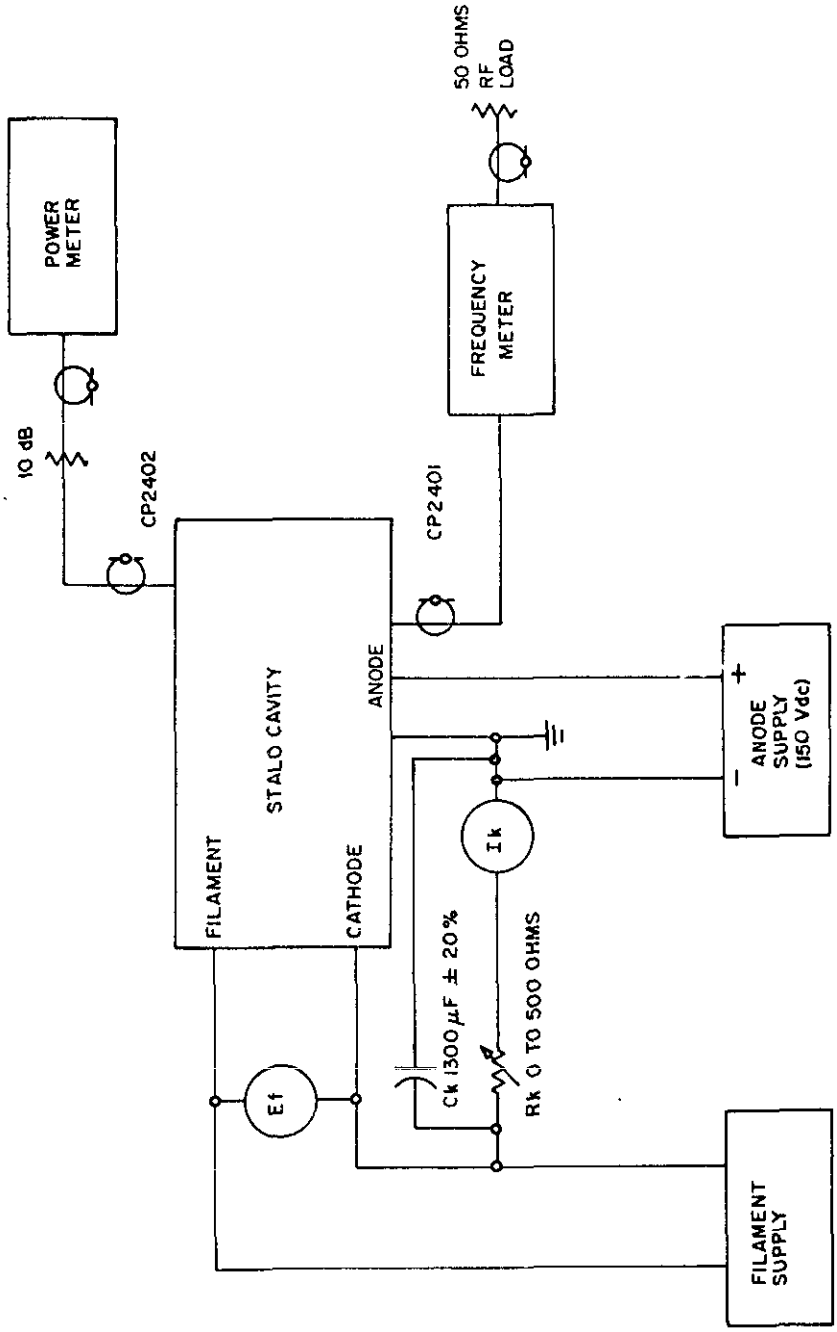


FIGURE 2. Power oscillation test circuit.

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